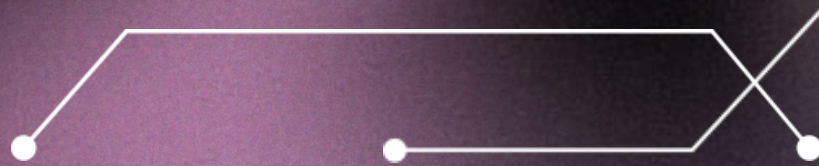




mmWave Radar Sensors

with Texas Instruments

Beyond Cameras: Accurate, Anonymous Human Activity Detection



Who are we?



Eddie Goodings
Board Wizard



Dylan Lin
Documentation Master



Edward Clark
Visualiser Visionary



Emmanuel Destin
Testing Emperor



Faik Zaim
Communication Captain

Our Documentation Website

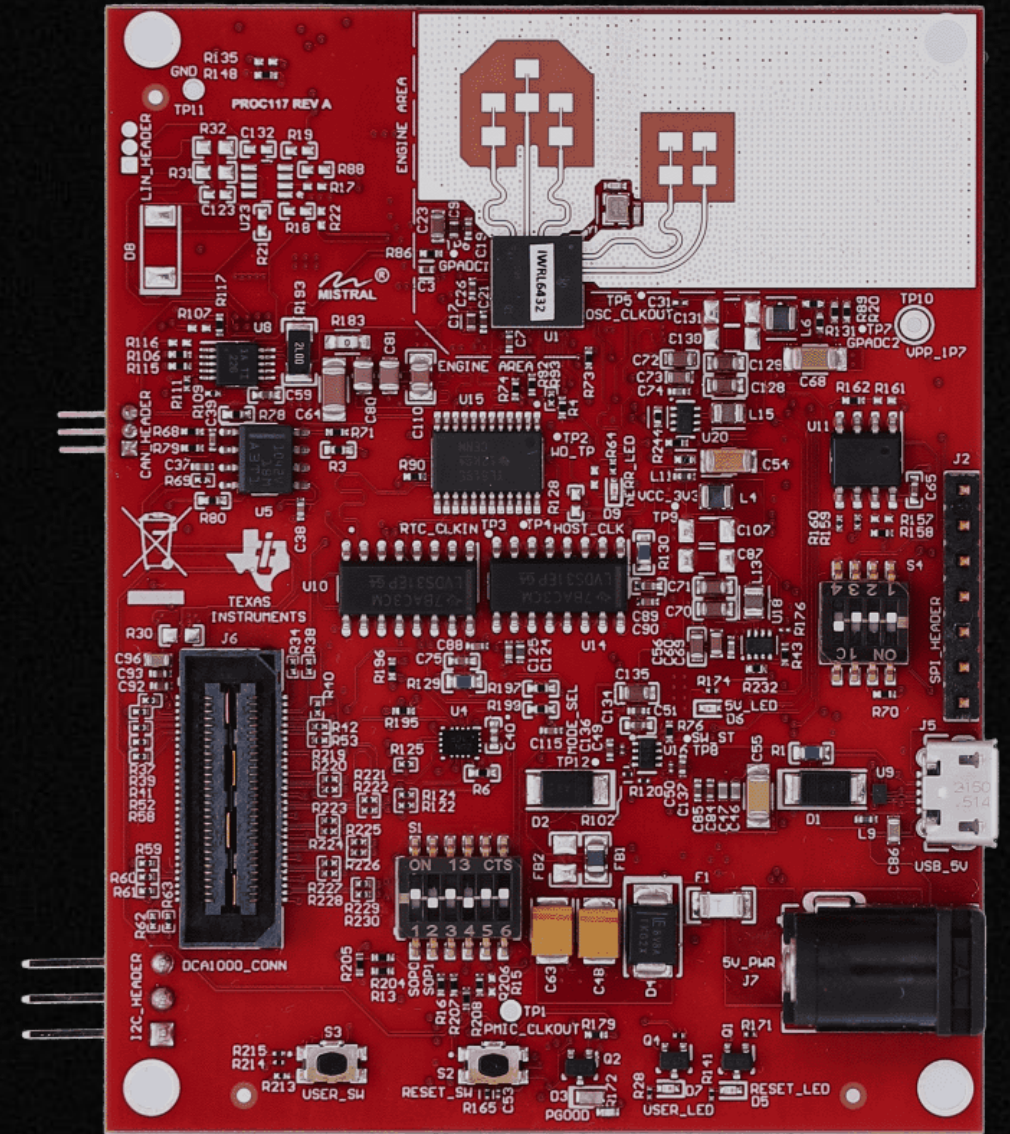


Develop a machine learning model that can detect different human postures.

Our brief

The Radar Board

- Detects **tiny movements** using millimetre-wave technology
- Emits invisible signals to sense movement **without relying on cameras**
- Works effectively in **all lighting conditions**, including complete darkness
- Low power consumption and **cost-effective**
- **Ensures privacy**, as it does not capture images or personally identifiable information.



User Stories



TI's Sales Managers

As a manager at Texas Instruments, I want to sell my radar chips by showing potential clients **example applications** that are relevant to them.



Hospital Patients

As a patient, I want to be safely monitored without compromising my **privacy**, so that I can receive immediate assistance if I fall or need help.

TI's Motivation



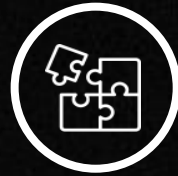
The posture model

We created a ML model that detected different human postures.



Example application

Our code will be published as an example use case for the hardware.



Observing the process

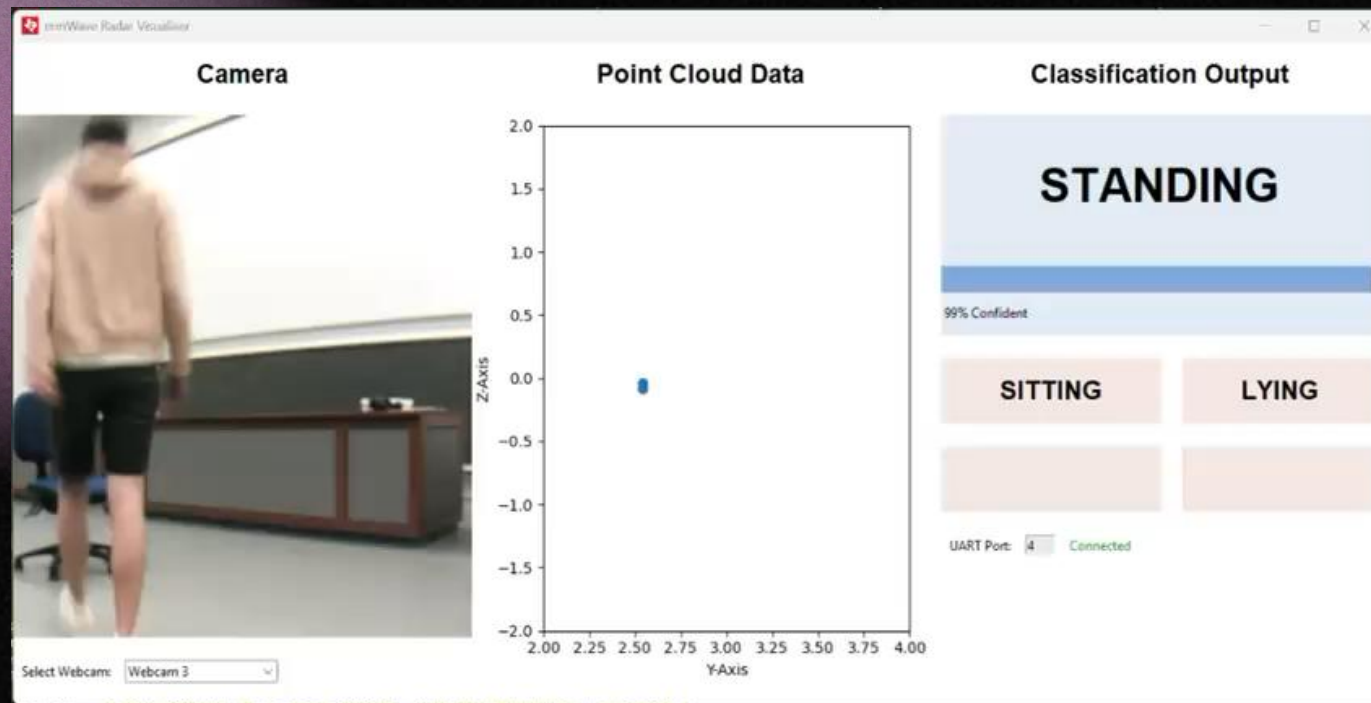
Our client was trying to identify all the development roadblocks.



Documentation feedback

We were beta testing documentation, finding mistakes before it's public.

Live Demonstration



Challenge:

Learning ML

None of us had studied machine learning or data before. There was a lot of information and resources thrown at us by our client.

Solution: We spent a significant portion of the MVP development stage learning, reading papers, and finding resources. All this information, as well as user instructions and data collection reports, have been published on our documentation website.

[View on GitHub](#)

mmWave Radar Sensor Project Documentation Site

A site full of our findings and documentation for the mmWave Radar Sensor Project with Texas Instruments

Installation Guide

1. These instructions are for Ubuntu 22.04 LTS either natively or running in WSL2.
2. Download the MMWAVE-L-SDK for xWRL6432 from <https://www.ti.com/tool/MMWAVE-L-SDK>. Install it in `C:/ti` on Windows or `~/ti` on Ubuntu.
3. Download SysConfig from <https://www.ti.com/tool/SYSCONFIG>. Install it in `C:/ti` on Windows or `~/ti` on Ubuntu.
4. Download the TI CLANG Compiler Toolchain from <https://www.ti.com/tool/TI-CLANG>. Install it in `C:/ti` on Windows or `~/ti` on Ubuntu.
5. Download and install Python 3.10 from <https://www.python.org/downloads/>. Make sure to add it to your path.
6. Make sure the Python package manager `pip` is installed.
7. Install the packages needed for the flashing tools with `pip install pyserial xmodem tqdm`.

Challenge:

A New Team Member

Emmanuel only joined the project after our MVP release. This meant that he missed out on all our research and learning time. He had a lot to catch up on!

Solution: Using our compiled research documentation, Texas Instruments' resources, and our help flashing the example application to the radar board, he got up to speed very quick.

 2024-mmWaveRadarSensors #39 ... 

Flash the "hello world" example to the radar board to learn the process

Priority: HIGH Story Points: 5 project setup

📌 Start of MVP Development Sprint Oct 17, 2024



destine2112 27/10/2024 22:39

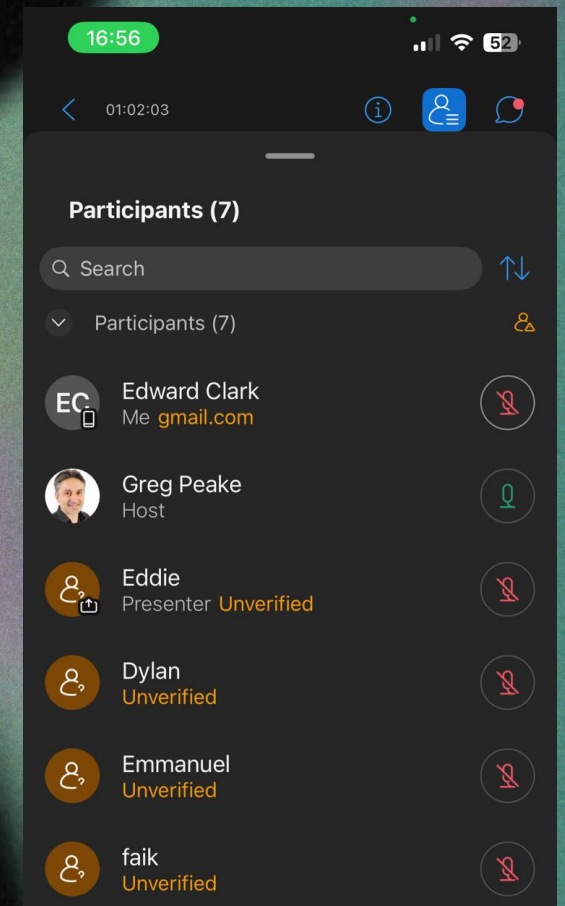
Hey guys, thanks for letting me joining your project. It seems rly interesting

Challenge:

Brand New Hardware

The reason Texas Instruments gave this project to us was to help them identify and remove all the issues that occur when developing for their new radar chips.

Solution: Our client was invaluable. We scheduled fortnightly meetings with him for the duration of the project. There was always something to discuss. We worked together on hardware issues, embedded C bugs, and weird ML model behaviour.

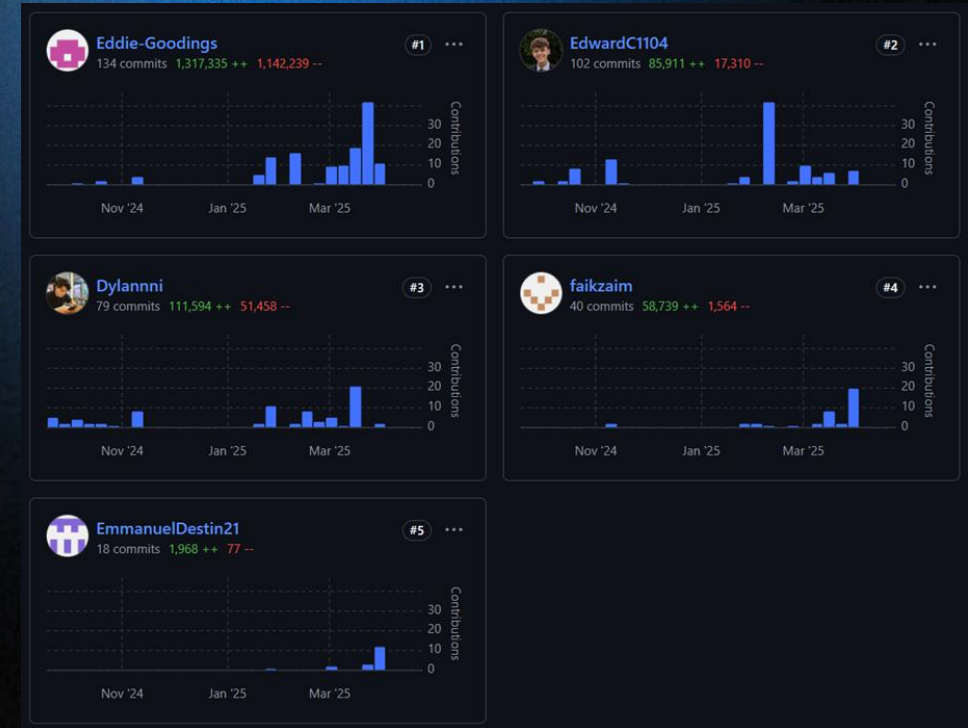


Challenge:

Splitting the Work

Understandably, it was tricky finding different sections of the project for us to each dive into at the beginning. There was so much to learn before real development could begin.

Solution: The project really felt like it launched in January. We met our client in person to discuss developing our own new software. We also had a team review where we agreed to split into three pairs each with our own development focus.

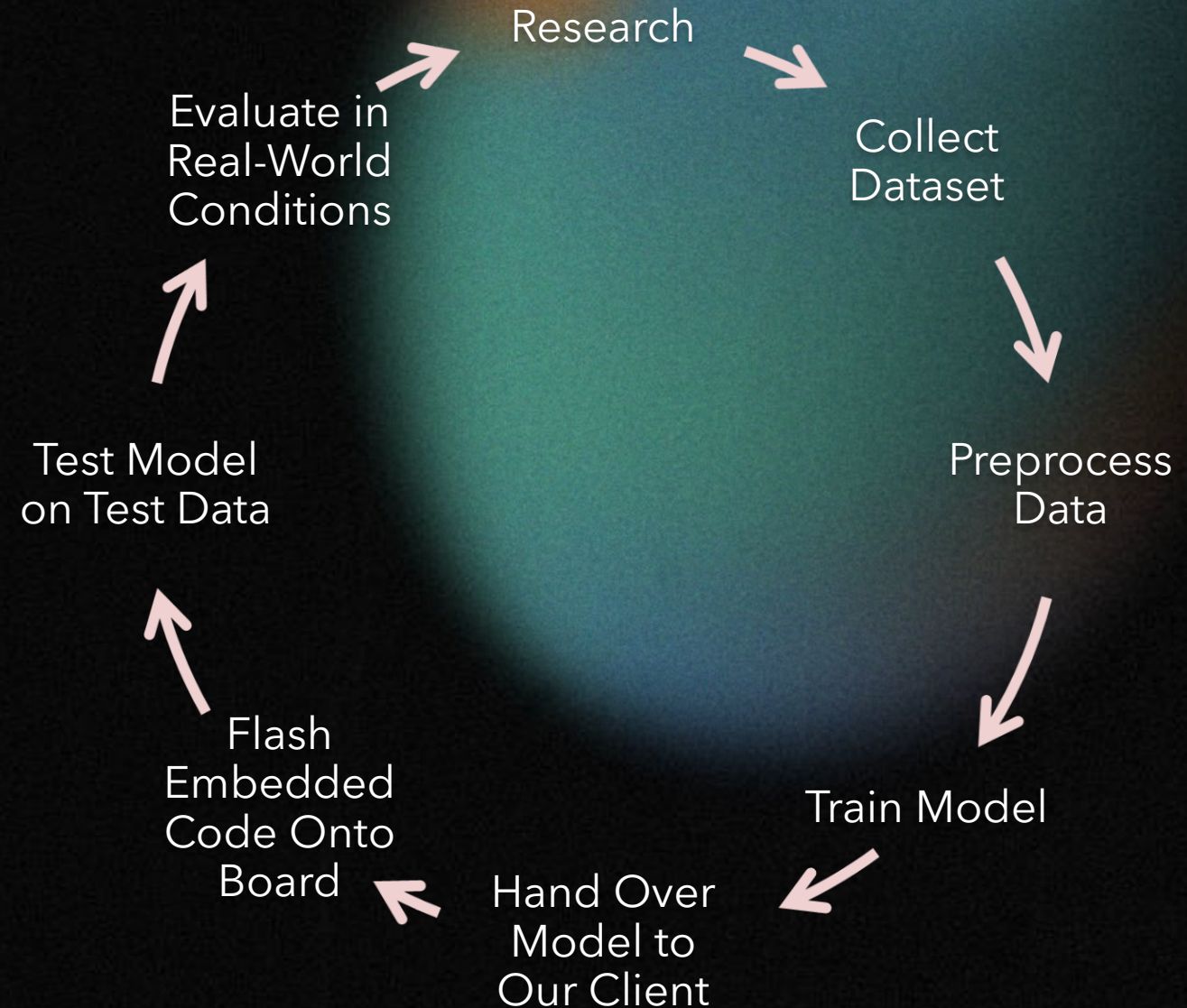


Our Process

Our development followed a clear cycle: collect data, train, release to TI, integrate, test, and evaluate.

Each model handover created a natural sprint, with built-in review periods while our client helped us flash the model onto the board.

This enforced an Agile flow of continuous development, feedback, and improvement.

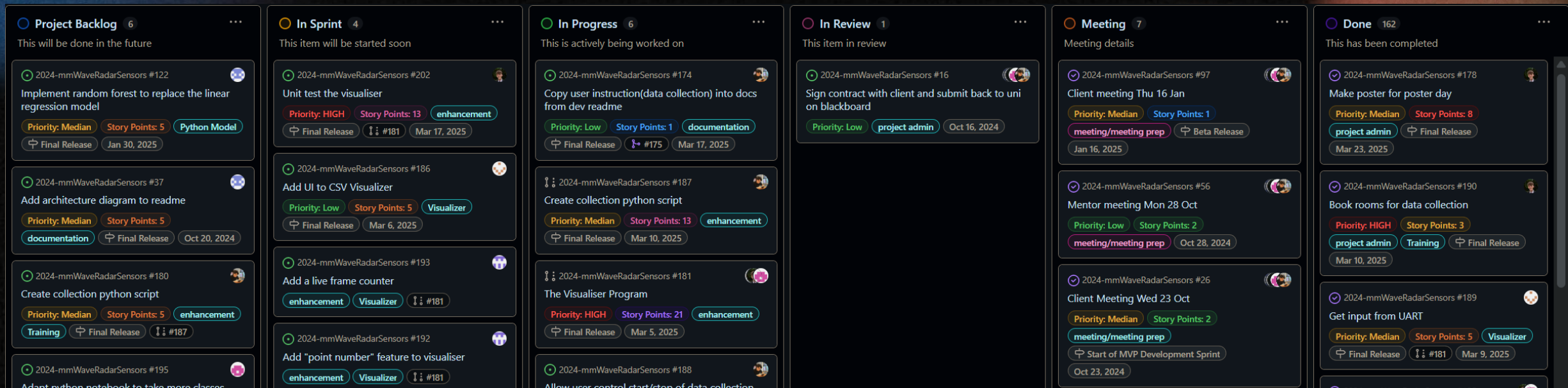
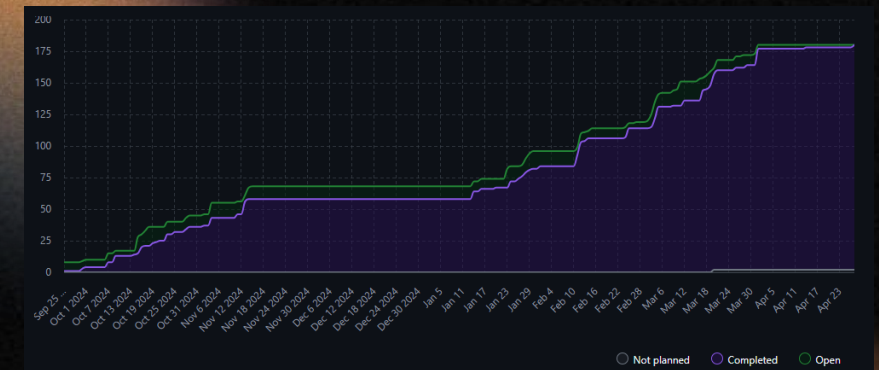


The Kanban Board

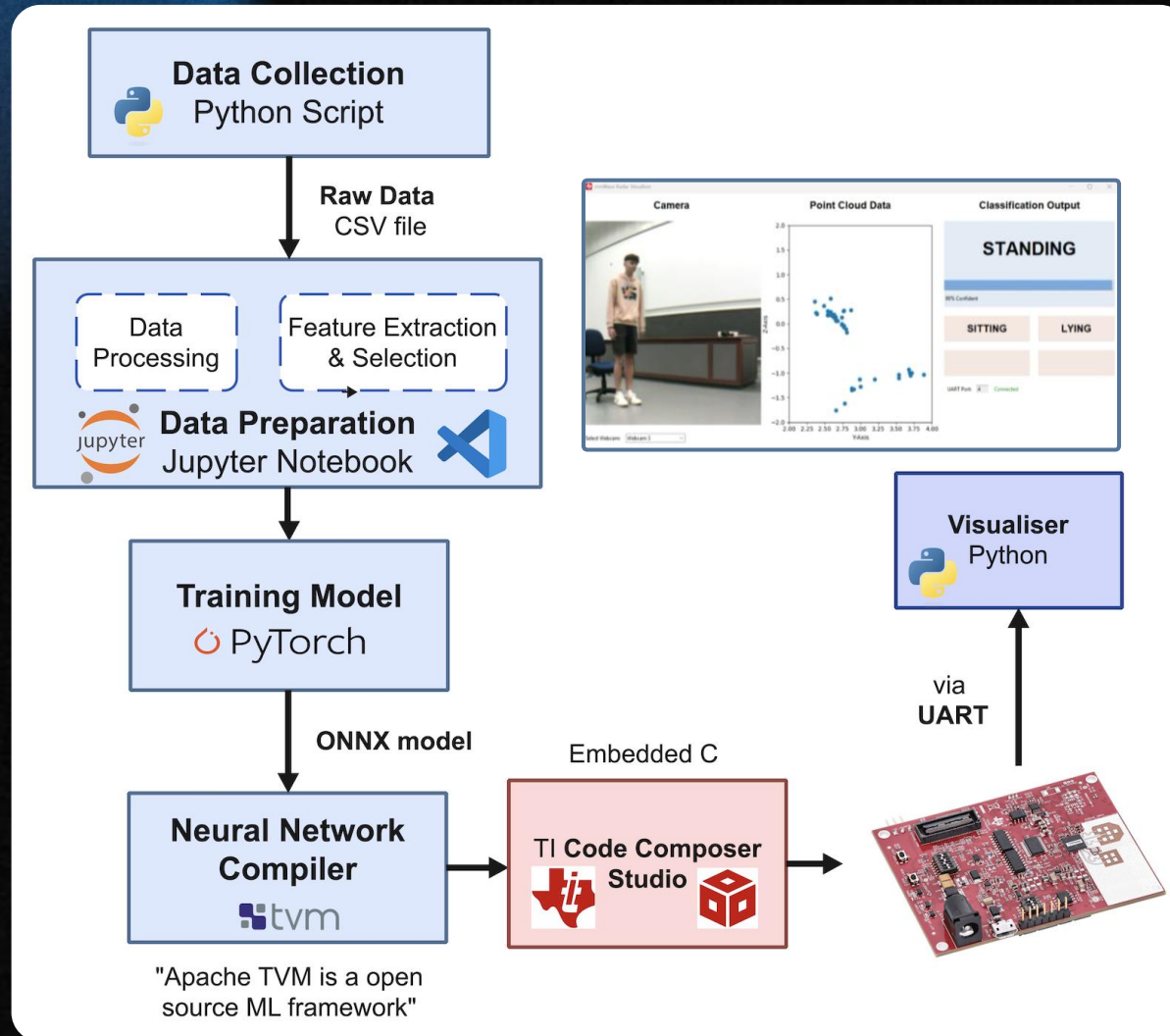
The Kanban board was an invaluable tool that drove the project forward at every stage.

We used Agile style categories (backlog, sprint, etc.) and labels which gave issues a milestone, priority, story point value, type, and more.

Burn up

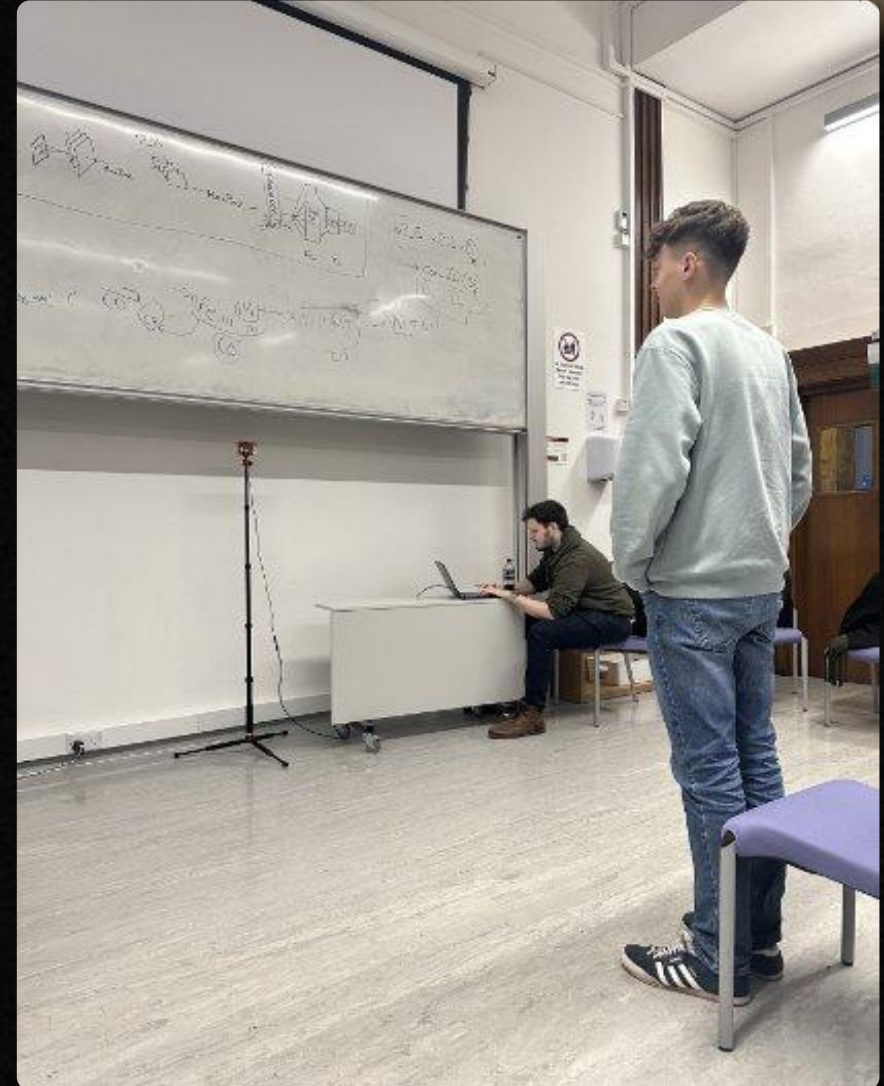


Architecture Diagram

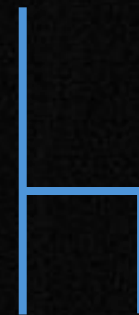
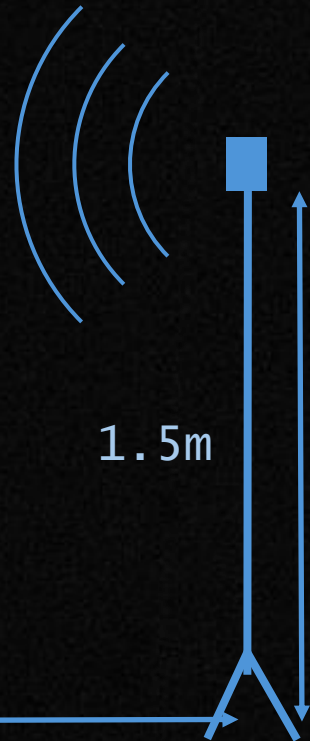
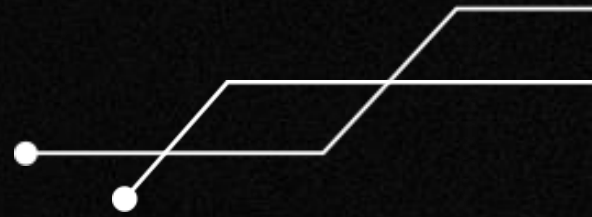
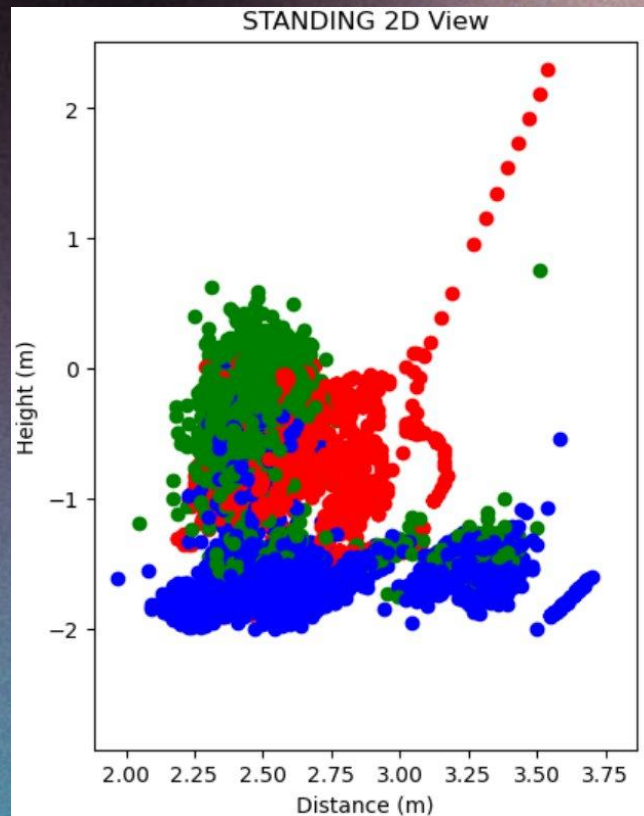
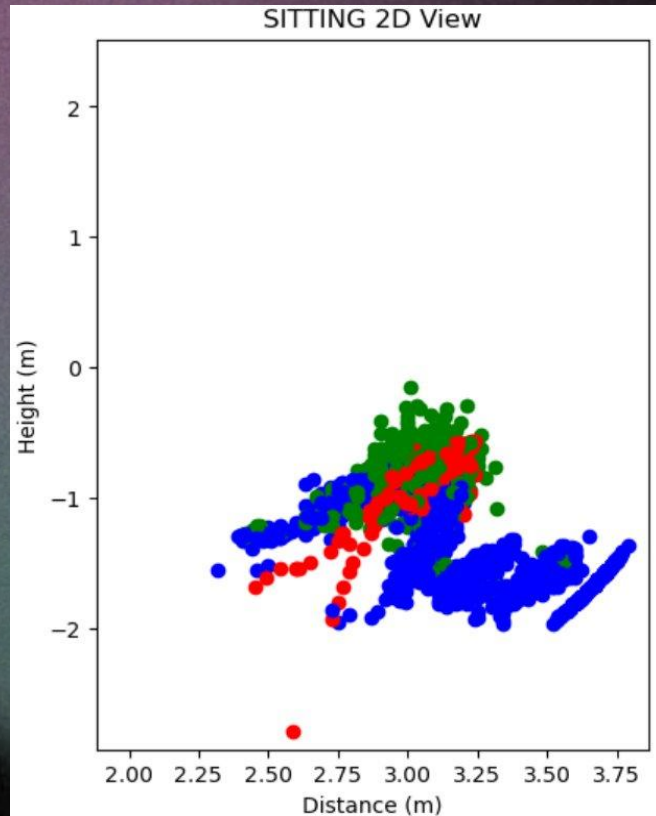


Introduction to data collection

- We completed **three sprints** which included data collection
- Each data collection iteration required:
 - Location and Setup planning
 - Collection
 - Post-processing
 - Documentation

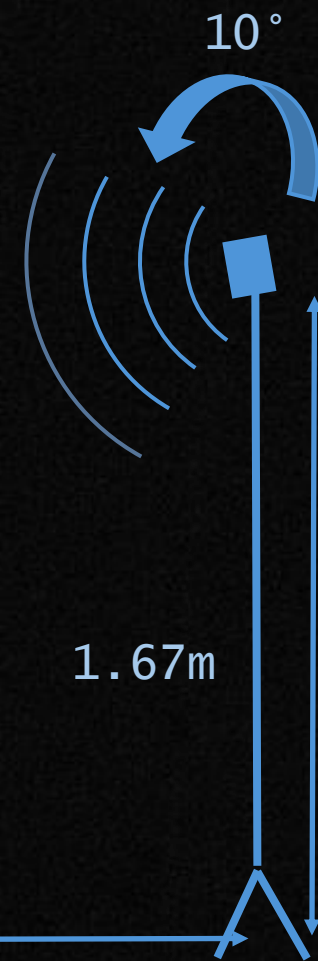
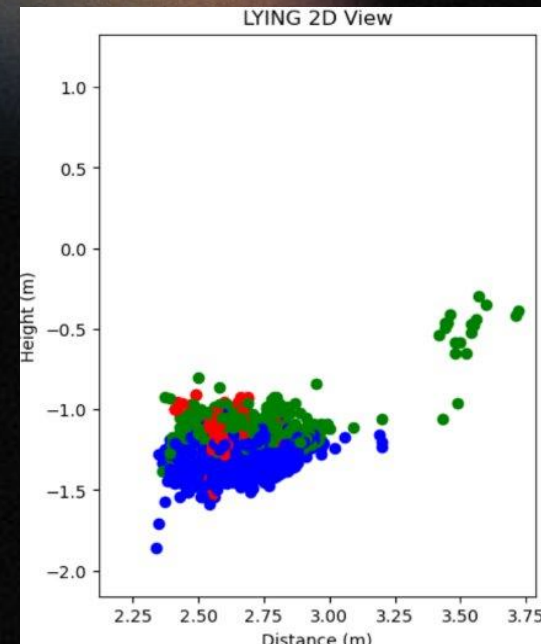
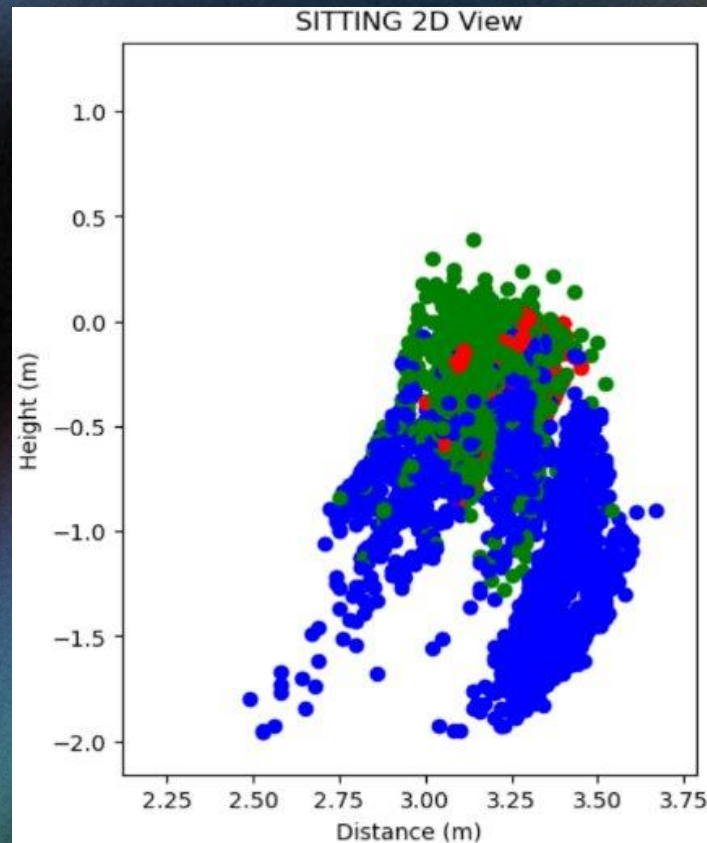
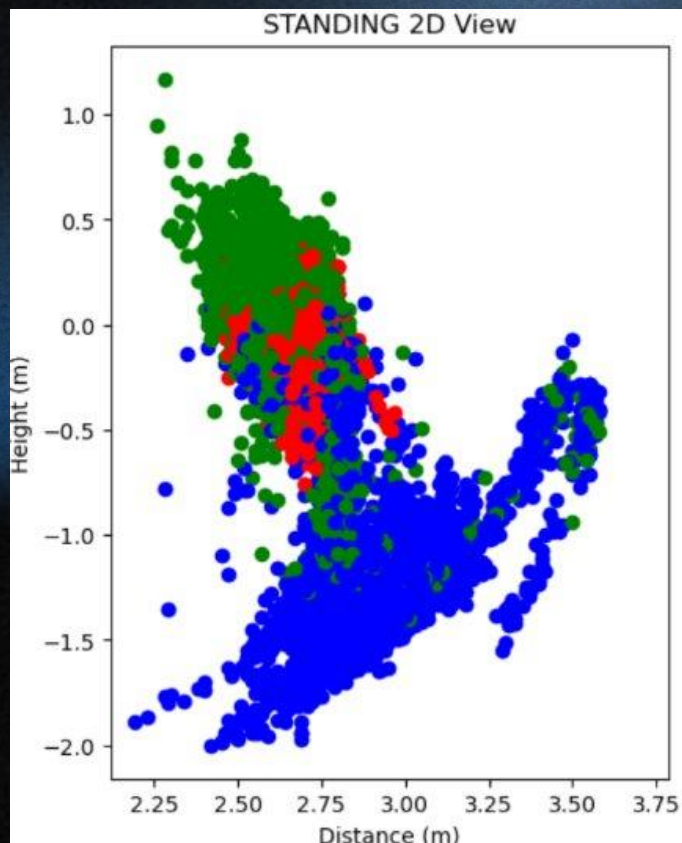


Collection 1: A Proof of Concept



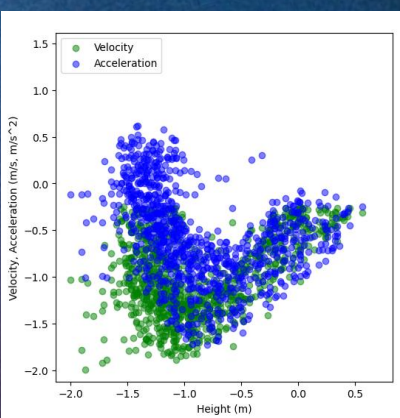
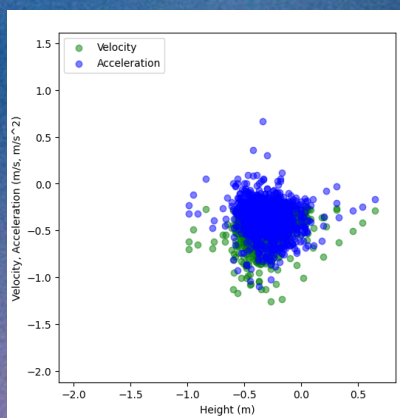
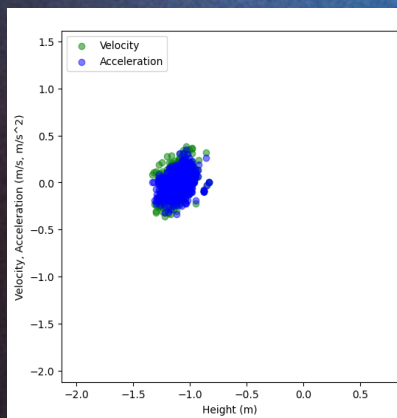
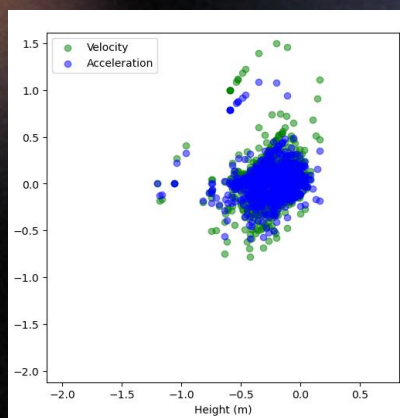
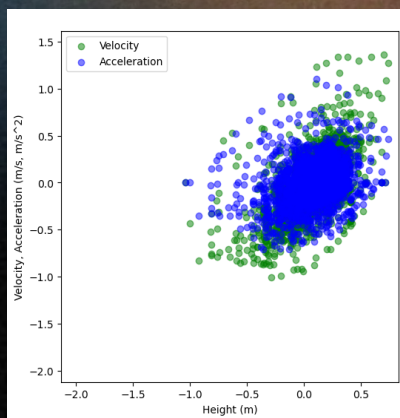
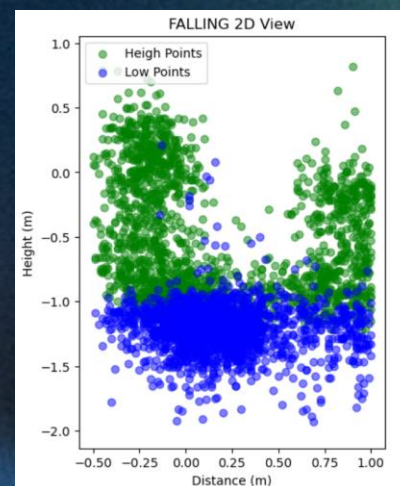
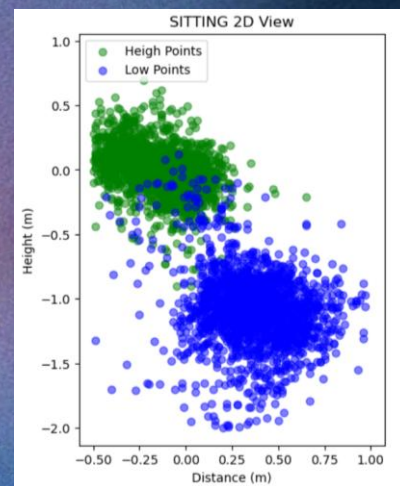
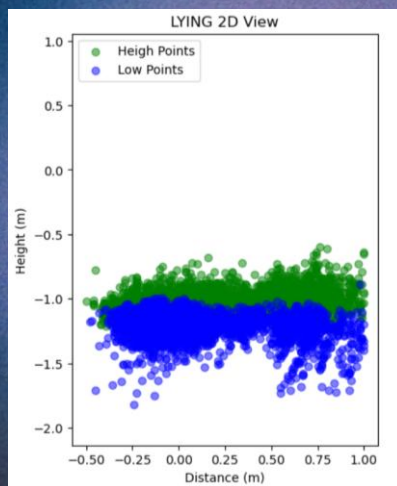
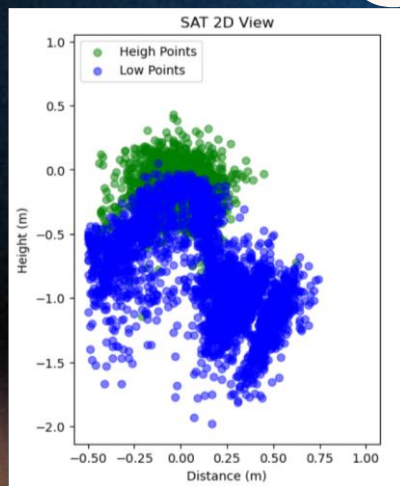
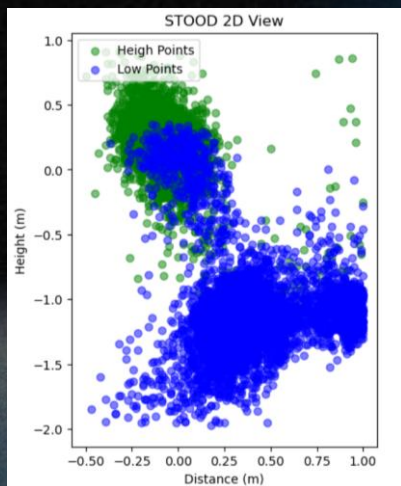
🙄 Only two graphs...

Collection 2: A test of Viability



😬 Oh dear there's another one...

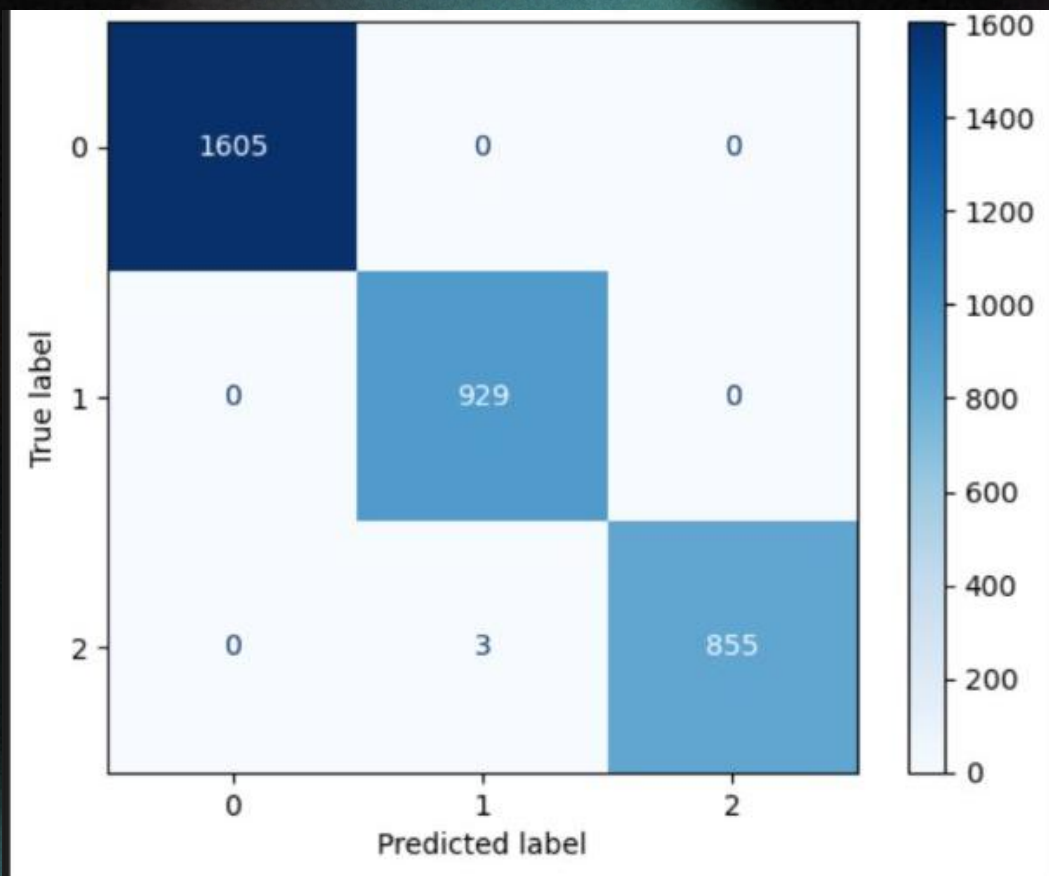
Collection 3: A Functioning Model



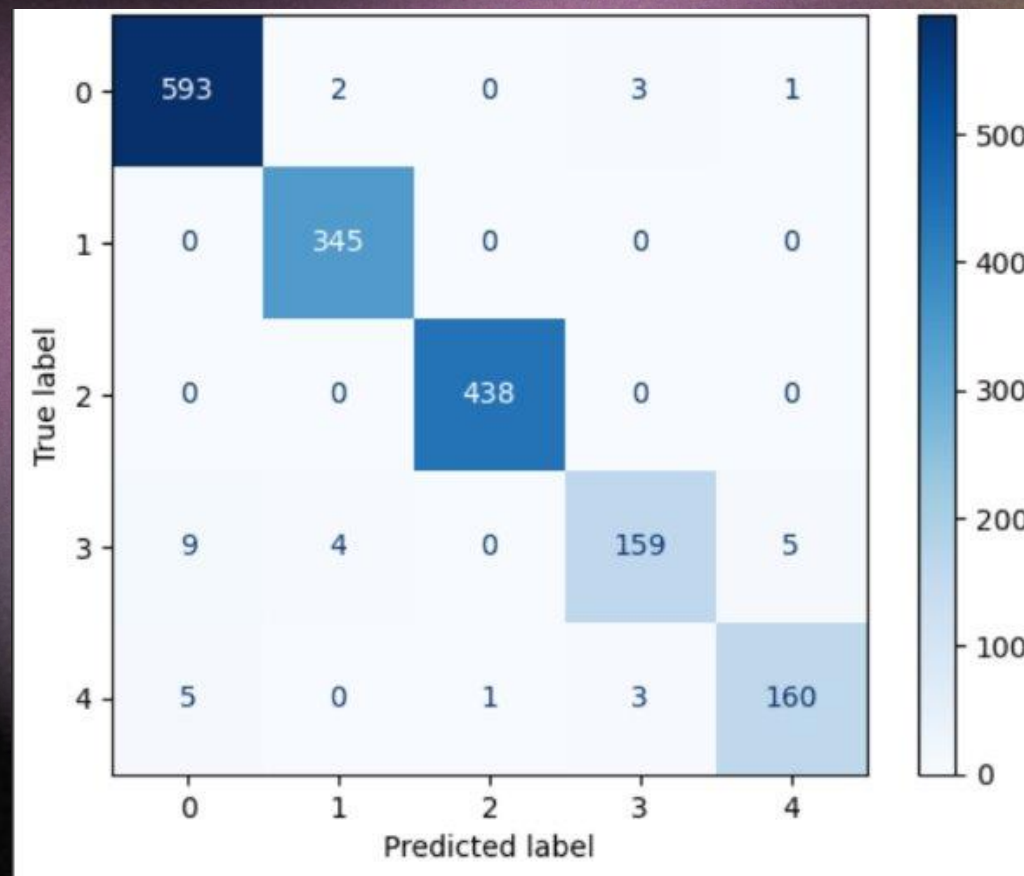
What is that???

Confusion Matrices

First Viable Model:



All Classes:



0: Stood 1: Sat 2: Led 3: Sitting 4: Falling

Summary of data collections

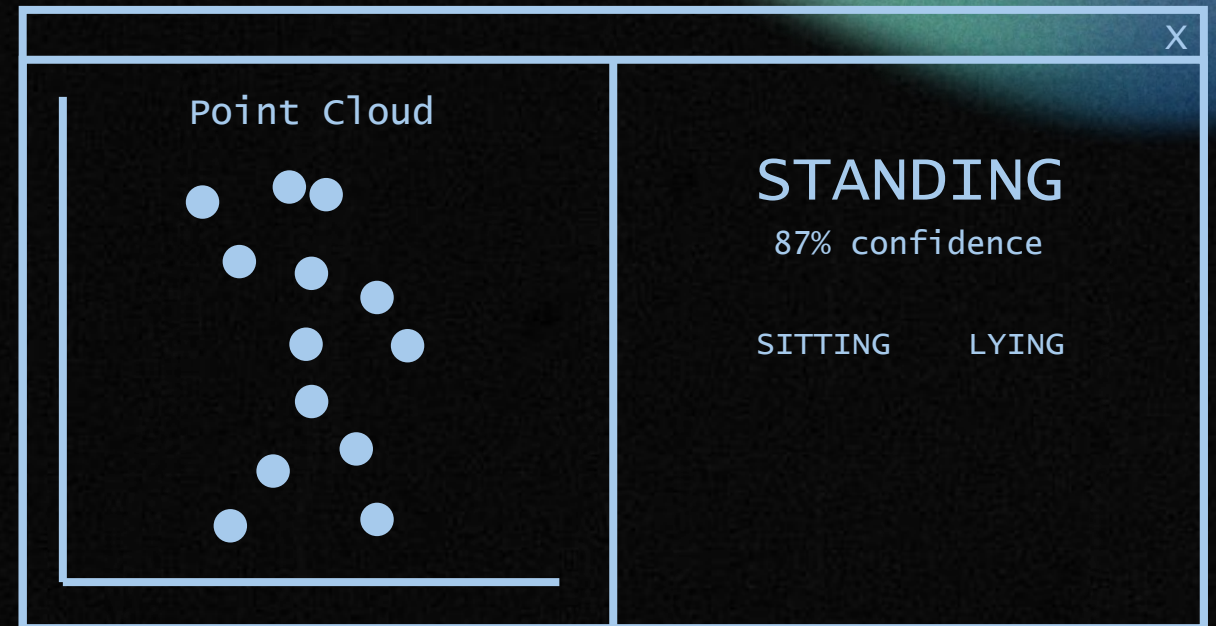
- **Improved Collection Parameters** at each stage to produce better data for our model
- **Changed how our model used the data** based on observations made during collection
- **Developed our own script** to aid in the collection of more complex poses



Visualiser

- It's tough to improve data or tweak the model when we can't even see what the board sees—so we built a visualiser to bring it all to life
- We listened to the user stories and pitched a visualiser feature to the client.
- He gave us feedback to include a camera view as a “source of truth” for developers, so we took that on board and added it.

- ☒ Figma design
- ☒ Point cloud data graph
- ☒ Link to radar board
- ☒ Model output UI
- ☐ Support for different models

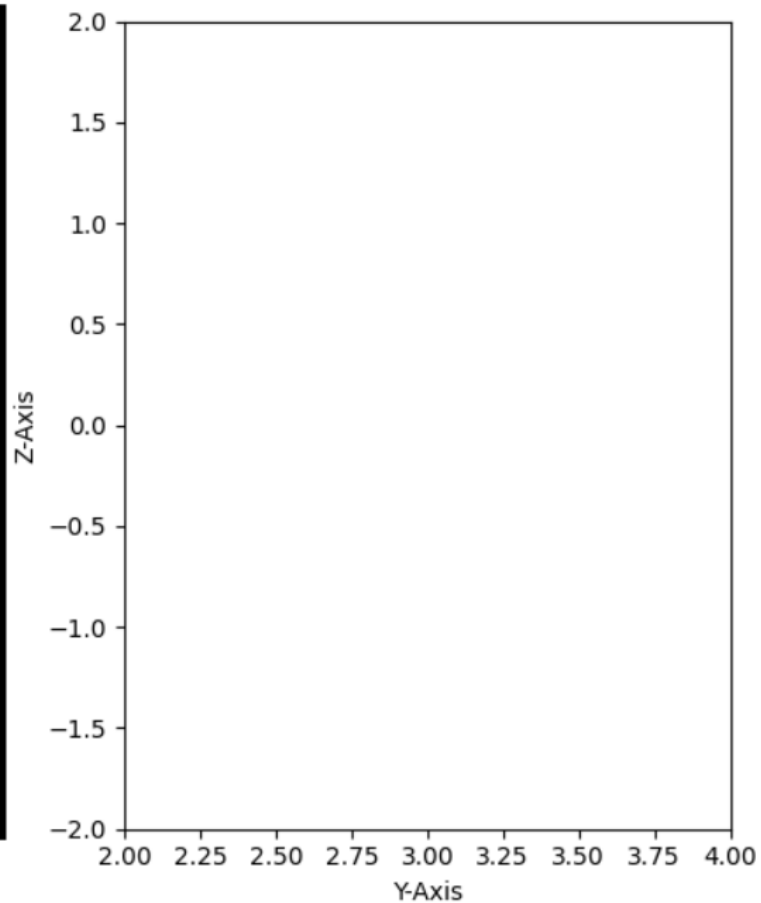


Camera



Select Webcam:

Point Cloud Data



Classification Output

STOOD

0% Confident

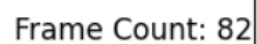
SAT

LYING

SITTING

FALLING

UART Port: Connected

[Open CSV File](#)

Continuous Integration

- CI runs our test suite.
- Challenge we faced (Jupyter Notebook and regular Python)

Triggered via push last month

EdwardC1104 pushed  60fd07c [main](#)

Status

Success

Total duration

18m 51s

continuous-integration.yml

on: push

 build-and-test

18m 43s

Continuous Deployment



Automatically
adds new
models



Searches for
and executes
every Jupyter
notebook



Searches for
onnx model files



Names them
based on the
directory
structure



Publishes them
as a GitHub
release

Continuous Deployment

Triggered via push last month

 EdwardC1104 pushed -o- 60fd07c `main`

Status

Success

Total duration

16m 9s

continuous-deployment.yml

on: push

✓ **run-notebooks**

15m 44s

✓ **create-release**

7s

ONNX Files v34

This release contains all the ONNX files generated by our models.

Assets 6

 [dry_wet.onnx](#)

 [grass_notgrass.onnx](#)

 [posture_absolute.onnx](#)

 [posture_relative.onnx](#)

 [Source code \(zip\)](#)

 [Source code \(tar.gz\)](#)

Feedback/Some user testing

- People thought the camera output was used alongside the radar to detect posture.
- We deactivated the camera view during poster day to avoid confusion.
- We got lots of positive feedback about how our technology can be used without comprising privacy.

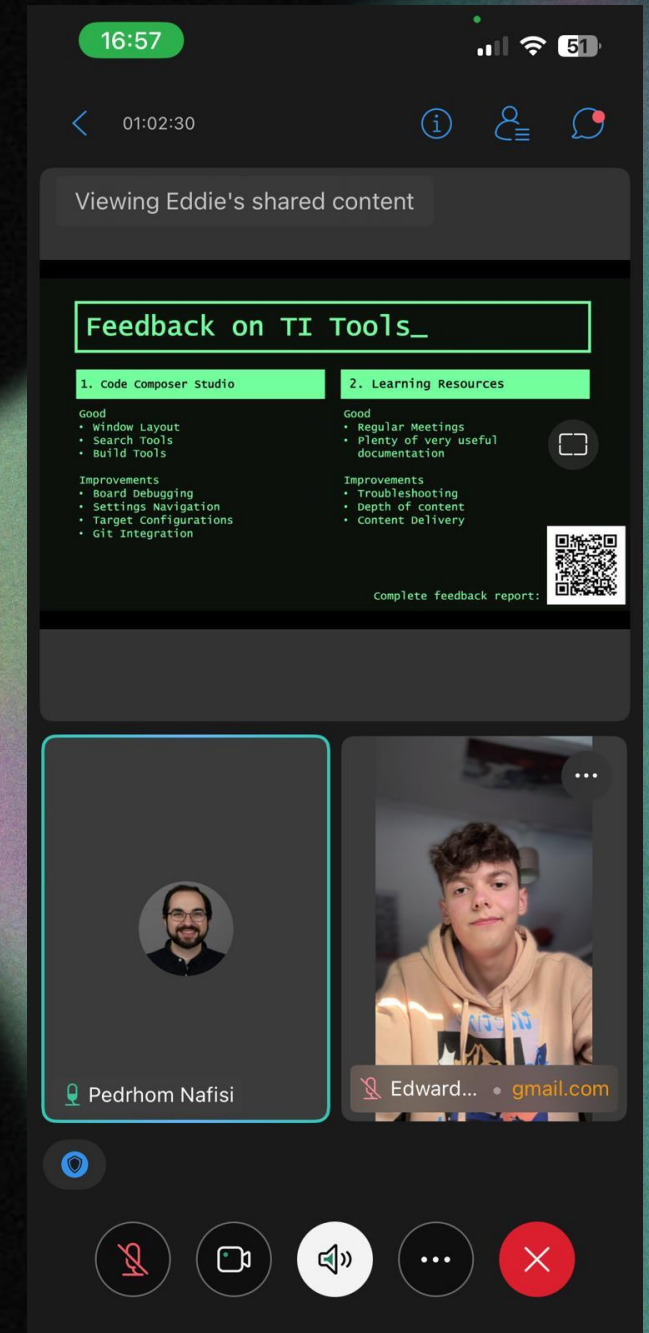


Handover Presentation

- The handover gave us a lot of great constructive criticism.
- An engineer from TI gave us tips on how we could improve model accuracy!
- Create a confidence threshold, only give an output if the confidence is above this threshold.
- Only switch the output given to a new posture if the model has detected it multiple times consecutively.

Team, thanks again for the presentation yesterday – well presented and well received. I didn't find the ppt in the bundle -can you pls send that as well ?

Regards, greg





Any Questions?

Documentation Website:



- ✓ Developed a ML model to identify human postures
 - ✓ Documented our application, data collection, and research
 - ✓ Worked with our client to overcome development hurdles
 - ✓ Shared our thoughts on new resources and documentation
 - ✓ Used Agile development effectively
 - ✓ Deployed CI and CD
 - ✓ Managed project challenges collaboratively as a team
- 